Investigación sobre osteopatía

INTEGRANTES:

- **Ayala Huaman Betsabe**
- **Espinoza Marin Melissa**
- **Fernández Silva Camila**
- **Herrera Delao Eilleen**
- **Leon Flores Yudit**
- **Valdivia Izquierdo Lucero**



PREGUNTA DE INVESTIGACIÓN

En **estudiantes** de primer año de medicina de la Universidad Tecnológica del Perú con **mialgia** asociada al **estrés académico** (P), ¿cuál es el efecto de la **terapia osteopática** (I), en comparación con la **ausencia de tratamiento** (C), para la **reducción de la intensidad del dolor musculoesquelético** (O)?



¿Qué es Osteopatía?

La osteopatía es una disciplina terapéutica basada en la manipulación manual de músculos, articulaciones, huesos y tejidos conectivos para restaurar el equilibrio funcional del cuerpo y estimular sus mecanismos naturales de autorregulación y autocuración, donde su enfoque es holístico, considerando que todas las estructuras del organismo están interconectadas y que una disfunción en una parte puede afectar otras áreas



LA OSTEOPATÍA PODRÍA AYUDAR A REGULAR LA RESPUESTA DEL ORGANISMO AL ESTRÉS Y FAVORECER LA RELAJACIÓN MEDIANTE TÉCNICAS MANUALES, AUNQUE TODAVÍA SE NECESITAN MÁS ESTUDIOS PARA CONFIRMAR SU EFICACIA.

Artículo 1

OBJETIVO DEL ESTUDIO

Resumir la evidencia clínica disponible sobre la eficacia y seguridad del tratamiento manipulativo osteopático (OMT) para diferentes condiciones de salud, a partir de revisiones sistemáticas y metaanálisis previos de ensayos clínicos aleatorizados.

POBLACIÓN Y MUESTRA

Estudios incluidos: 9 revisiones sistemáticas (55 ECA; n = 3,740 pacientes).

Áreas de estudio: Dolor musculoesquelético (lumbalgia, cervicalgia), patologías pediátricas, cefaleas y trastornos viscerales (SII).

INFORMACIÓN TÉCNICA DEL ARTÍCULO:

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PMID: 35414546

Tipo: Revisión general de revisiones sistemáticas y metaanálisis

CUARTIL Y FACTOR DE IMPACTO

Revista: Journal of Osteopathic Medicine

Indexación; Web of Science (ESCI) y Scoopus

Factor de Impacto (JIF): 1,3

Posicionamiento en Q2 en JCR

Pocisionamiento en Q1 en SCImago / Scopus

PUBLICACIÓN: Abril 2022



Open access Original research

BMJ Open

Eficacia y seguridad del tratamiento manipulativo osteopático: una visión general de las revisiones sistemáticas

Efficacy and safety of osteopathic manipulative treatment: an overview of systematic reviews

Donatella Bagagiolo ,[1] Debora Rosa,[2] Francesca Borrelli[3]

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[1]Research Department, Scuola Superiore di Osteopatia Italiana, Turin, Italy
[2]Department of Cardiovascular, Neural and Metabolic Sciences, IRCCS Istituto Auxologico Italiano, Milan, Italy
[3]Department of Pharmacy, School of Medicine and Surgery, University of Naples Federico II, Naples, Italy

Correspondence to
Dr Donatella Bagagiolo, donatella.bagagiolo@gmail.com

ABSTRACT

Objective To summarise the available clinical evidence on the efficacy and safety of osteopathic manipulative treatment (OMT) for different conditions.

Design Overview of systematic reviews (SRs) and meta-analyses (MAs). PROSPERO CRD42020170983.

Data sources An electronic search was performed using seven databases: PubMed, EMBASE, CINAHL, Scopus, JBI, Prospero and Cochrane Library, from their inception until November 2021.

Eligibility criteria for selecting studies SRs and MAs of randomised controlled trials evaluating the efficacy and safety of OMT for any condition were included.

Data extraction and synthesis The data were independently extracted by two authors. The AMSTAR-2 tool was used to assess the methodological quality of the SRs and MAs. The overview was conducted and reported according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses statement.

Results The literature search revealed nine SRs or MAs conducted between 2013 and 2020 with 55 primary trials involving 3740 participants. The SRs reported a wide range of conditions including acute and chronic non-specific low back pain (NSLBP, four SRs), chronic non-specific neck pain(CNSNP, one SR), chronic non-cancer pain (CNCP, one SR), paediatric (one SR), neurological (primary headache, one SR) and irritable bowel syndrome (IBS, one SR). Although with a different effect size and quality of evidence, MAs reported that OMT is more effective than comparators in reducing pain and improving functional status in acute/chronic NSLBP, CNSNP and CNCP. Due to small sample size, presence of conflicting results and high heterogeneity, questionable evidence existed on OMT efficacy for paediatric conditions, primary headache and IBS.

No adverse events were reported in most SRs. According to AMSTAR-2, the methodological quality of the included SRs was rated low or critically low.

Conclusion Based on the currently available SRs and MAs, promising evidence suggests the possible effectiveness of OMT for musculoskeletal disorders. Limited and inconclusive evidence occurs for paediatric conditions, primary headache and IBS. Further well-conducted SRs and MAs are needed to confirm and extend the efficacy and safety of OMT.

Strengths and limitations of this study

- This systematic overview included a comprehensive literature search for evidence on the efficacy and safety of osteopathic manipulative treatment (OMT) for any condition.
- The present overview was conducted according to the Cochrane Handbook for the Systematic Review of Interventions and reported following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines.
- The inclusion criteria were restricted to systematic reviews and meta-analyses of randomised controlled trials that included patients with any conditions.
- Since only randomised controlled trials in which OMT was performed by osteopathic physicians or osteopaths were included, some relevant systematic reviews could have been missed.
- The quality of the evidence from the included systematic reviews and meta-analyses was assessed according to the AMSTAR-2 tool.

INTRODUCTION

Osteopathic medicine, depending on different legal and regulatory structures around the world, is a medical profession (eg, USA), an allied health profession (eg, UK) or a part of complementary and alternative medicine[1] (CAM; eg, Italy or France), developed by Andrew Taylor Still in the late 1800s in the Midwestern USA.[2] This therapy is based on the principle that the structure (anatomy) and function (physiology) of the individual's body are closely integrated and that a person's well-being depends on the balance of neurological, musculoskeletal and visceral structures.[3]

Osteopathic medicine is provided on almost every continent, and in 2020, a survey estimated that 106861 osteopathic practitioners provide osteopathic care worldwide in 46 countries.[4]

Osteopathic medicine plays an important role primarily in musculoskeletal healthcare. A recent survey conducted in Switzerland[5] on a sample of 1144 patients showed that over 80% of patients had requested an osteopathic consultation for musculoskeletal pain [mainly

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LIMITACIONES

Calidad metodológica de las RS clasificada como baja a críticamente baja (AMSTAR-2)

Alto riesgo de sesgo por falta de cegamiento en los ECA primarios

Alta heterogeneidad en las intervenciones y técnicas de TMO

Presencia de variables intervinientes: Heterogeneidad de controles e insuficiente reporte de eventos adversos

Table 3 Overall traffic light evidence for OMT efficacy

Conditions	First author, year	Overall traffic light evidence
Musculoskeletal conditions		
1.ANSLBP/CNSLBP		
Pain	Franke, 2014[25]	
	Dal Farra, 2020[27]	
Functional status	Franke, 2014[25]	
	Dal Farra, 2020[27]	
2.CNSLBP		
Pain	Franke, 2014[25]	
	Dal Farra, 2020[27]	
Functional status	Franke, 2014[25]	
	Dal Farra, 2020[27]	
3.NSLBP in pregnancy		
Pain	Franke, 2014[25]	
	Franke, 2017[28]	
Functional status	Franke, 2014[25]	
	Franke, 2017[28]	
4.NSLBP in PP		
Pain	Franke, 2014[25]	
	Franke, 2017[28]	
Functional status	Franke, 2014[25]	
	Franke, 2017[28]	
5.LBP with sciatica		
Pain	De Oliveira Meirelles, 2013[21]	
6.LBP with menopausal symptoms		
Pain	De Oliveira Meirelles, 2013[21]	
7.CNSNP		
Pain	Franke, 2015[29]	
Functional status	Franke, 2015[29]	
8.CNCP		
Pain	Rehmani, 2020[30]	
Disability	Rehman, 2020[18]	
Quality of life	Rehman, 2020[18]	
Paediatric conditions		
Outcomes for different conditions *	Posadzky, 2013[22]	
Neurological conditions		
Outcomes for migraine and tension-type headache†	Carnitelli, 2017[31]	
Visceral conditions		
Outcomes for IBS‡	Muller, 2014[32]	

PRINCIPALES CONCLUSIONES

Trastornos Musculoesqueléticos

El TMO demuestra efectividad clínica en dolor lumbar no específico y cervicalgia, siendo una terapia complementaria prometedora

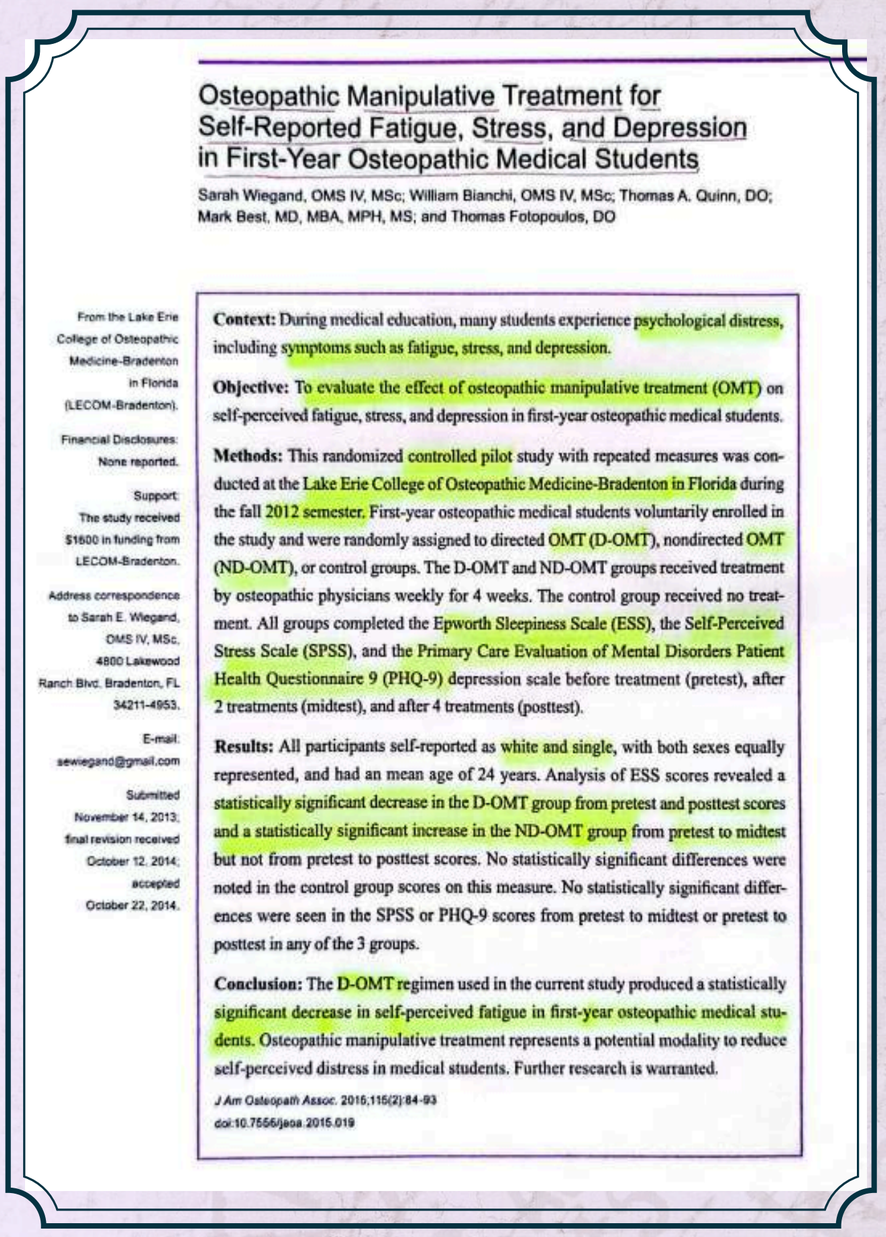
Ámbito No Musculoesquelético

Evidencia inconclusa o limitada para pediatría, cefaleas y trastornos viscerales. Se requieren ECA con mayor rigor.

Perfil de Seguridad

El TMO es una intervención segura sin reportes de eventos adversos graves; se observan únicamente efectos leves transitorios.

Artículo 2



Osteopathic Manipulative Treatment for Self-Reported Fatigue, Stress, and Depression in First-Year Osteopathic Medical Students

Sarah Wiegand, OMS IV, MSc; William Bianchi, OMS IV, MSc; Thomas A. Quinn, DO; Mark Best, MD, MBA, MPH, MS; and Thomas Fotopoulos, DO

From the Lake Erie College of Osteopathic Medicine-Bradenton in Florida (LECOM-Bradenton).

Financial Disclosures: None reported.

Support: The study received $1600 in funding from LECOM-Bradenton.

Address correspondence to Sarah E. Wiegand, OMS IV, MSc, 4800 Lakewood Ranch Blvd. Bradenton, FL 34211-4953.

E-mail: sewiegand@gmail.com

Submitted November 14, 2013; final revision received October 12, 2014; accepted October 22, 2014.

Context: During medical education, many students experience psychological distress, including symptoms such as fatigue, stress, and depression.

Objective: To evaluate the effect of osteopathic manipulative treatment (OMT) on self-perceived fatigue, stress, and depression in first-year osteopathic medical students.

Methods: This randomized controlled pilot study with repeated measures was conducted at the Lake Erie College of Osteopathic Medicine-Bradenton in Florida during the fall 2012 semester. First-year osteopathic medical students voluntarily enrolled in the study and were randomly assigned to directed OMT (D-OMT), nondirected OMT (ND-OMT), or control groups. The D-OMT and ND-OMT groups received treatment by osteopathic physicians weekly for 4 weeks. The control group received no treatment. All groups completed the Epworth Sleepiness Scale (ESS), the Self-Perceived Stress Scale (SPSS), and the Primary Care Evaluation of Mental Disorders Patient Health Questionnaire 9 (PHQ-9) depression scale before treatment (pretest), after 2 treatments (midtest), and after 4 treatments (posttest).

Results: All participants self-reported as white and single, with both sexes equally represented, and had an mean age of 24 years. Analysis of ESS scores revealed a statistically significant decrease in the D-OMT group from pretest and posttest scores and a statistically significant increase in the ND-OMT group from pretest to midtest but not from pretest to posttest scores. No statistically significant differences were noted in the control group scores on this measure. No statistically significant differences were seen in the SPSS or PHQ-9 scores from pretest to midtest or pretest to posttest in any of the 3 groups.

Conclusion: The D-OMT regimen used in the current study produced a statistically significant decrease in self-perceived fatigue in first-year osteopathic medical students. Osteopathic manipulative treatment represents a potential modality to reduce self-perceived distress in medical students. Further research is warranted.

J Am Osteopath Assoc. 2015;115(2):84-93
doi:10.7556/jaoa.2015.019

RELEVANCIA INTERNACIONAL

- Registrada en PubMed
- Clasificada como Randomized Controlled Trial

INFORMACIÓN TÉCNICA DEL ARTÍCULO:

DOI: 10.7556/jaoa.2015.019

PMID: 25637614

Tipo: ensayo controlado aleatorizado piloto con medidas repetidas.

AÑO DE PUBLICACIÓN: 2015

CUARTIL Y FACTOR DE IMPACTO

- Corresponden al Journal of Osteopathic Medicine
- Indexado en Web of Science/ESCI y Scopus
- JIF 1,3
- Posicionamiento Q2 en JCR
- Q1 según categoría en SCImago



ORGANIZACIÓN INTERNA DEL ARTÍCULO:

OBJETIVO DEL ESTUDIO

Evaluar el efecto del TMO sobre la fatiga, el estrés y la depresión auto percibidos en estudiantes de medicina osteopática de primer año.

POBLACIÓN Y MUESTRA

- Población: 104 estudiantes.
- Muestra: Al inicio eran 30 estudiantes, pero solo culminaron 28.

DURACIÓN Y DISTRIBUCIÓN

El proyecto tuvo una duración de 1 mes, divididos en 3 grupos, cada sesión tuvo una duración de 20 minutos.

LIMITACIONES

- Tamaño de muestra pequeño
- Ausencia de análisis estadístico por grupos
- Sesgo en la selección de la población
- Presencia de variables intervinientes: Programación de exámenes durante la experimentación

Escala: Epworth Sleepiness Scale
- 10-5 Fatiga baja diurna
- 6-10 Fatiga alta diurna

Escala: Estrés percibido
- 0-13 Estrés bajo
- 14-26 Estrés moderado
- 27-40 Estrés alto

Table.
Descriptive Analysis of Outcome Scores in Students Assessed for Psychological Distress (N=28)[a]

Group	Prestest Mean (SD)	Prestest Median (25%, 75%)	Midtest Mean (SD)	Midtest Median (25%, 75%)	Posttest Mean (SD)	Posttest Median (25%, 75%)
Epworth Sleepiness Scale[a] (Escala de sueño)						
D-OMT (n=10)	8.7 (4.4)	8 (5, 12)	6.3 (3.4)[b]	7 (3, 8)	5.9 (4.0)[b]	5 (2, 9)
ND-OMT (n=9)	6.2 (2.9)	7 (3.75, 8)	6.8 (3.5)	6 (4.75, 9)	6.3 (2.7)	7 (5.5, 7.25)
Control (n=9)	7.4 (3.1)	8 (4.75, 10)	8.6 (3.7)	8 (5.5, 11)	8.7 (3.5)	8 (6, 10.75)
Self-Perceived Stress Scale[c] (Escala de estrés)						
D-OMT (n=10)	25.3 (7.8)	24.5 (19, 31)	23.5 (5.8)	22.0 (19, 28)	24.9 (5.6)	23.5 (22, 28)
ND-OMT (n=9)	21.8 (8.1)	18.0 (16.75, 29.25)	22.0 (6.5)	20.0 (17.5, 27.5)	21.6 (7.0)	19.0 (16.75, 27.75)
Control (n=9)	22.0 (6.7)	22.0 (17, 25.75)	24.3 (8.3)	27.0 (18.25, 29.25)	22.1 (8.8)	22.0 (19.25, 25)
PHQ-9[d]						
D-OMT (n=10)	4.4 (3.3)	4.5 (2, 5)	3.0 (2.6)[b]	3.0 (1, 4)	3.4 (2.9)[b]	3.0 (2, 5) (Disminuyó)
ND-OMT (n=9)	4.7 (2.9)	4.0 (2, 7.25)	4.4 (3.4)	4.0 (1.75, 7)	6.3 (2.7)	7.0 (5.5, 7.25) (Aumentó)
Control (n=9)	3.8 (3.2)	3.0 (2, 4.5)	4.8 (3.9)	4.0 (1.75, 7.5)	4.3 (3.0)	4.0 (3.75, 5.5) (Disminuyó y se mantuvo)

[a] The Epworth Sleepiness Scale consisted of 8 questions scored from 0 to 3; a score of 10 or greater indicated fatigue.
[b] Statistically significant difference from pretest.
[c] The Self-Perceived Stress Scale contained 14 questions scored from 0 to 4; a score of 25 or greater indicated high stress.
[d] The Primary Care Evaluation of Mental Disorders Patient Health Questionnaire 9 (PHQ-9) depression scale contained 9 questions scored from 0 to 3. Total scores were interpreted as follows: 0-4, minimal depression; 5-9, mild depression; 10-14, moderate depression; 15-19, moderately severe depression; and 20-27, severe depression.[18]

Abbreviations: D-OMT, directed osteopathic manipulative treatment (OMT); ND-OMT, nondirected OMT.

Medidor del nivel de fatiga

Medidor del nivel de estrés

Medidor del nivel de depresión

Tratamiento directo

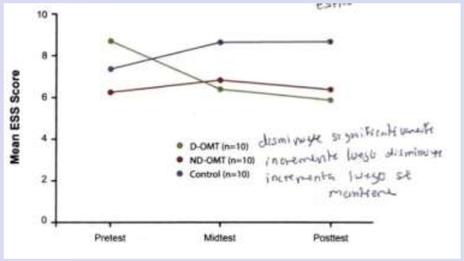
Tratamiento indirecto

Escala PHQ-9
- 10-4 mínima
- 5-9 leve

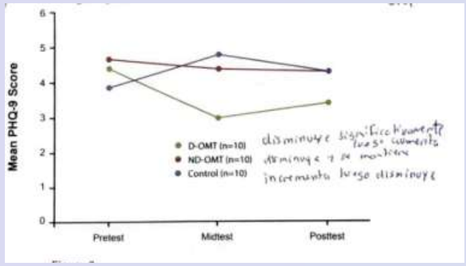
Se recomienda observación y tratamiento clínico

Estudio ... rolled in their first semester ...

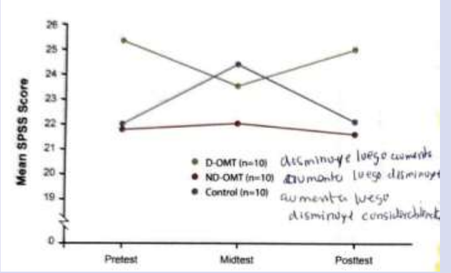
PRINCIPALES CONCLUSIONES



El análisis de las puntuaciones de la ESS reveló, una disminución estadísticamente significativa en el GRUPO D-TMO entre la prueba y la postprueba.

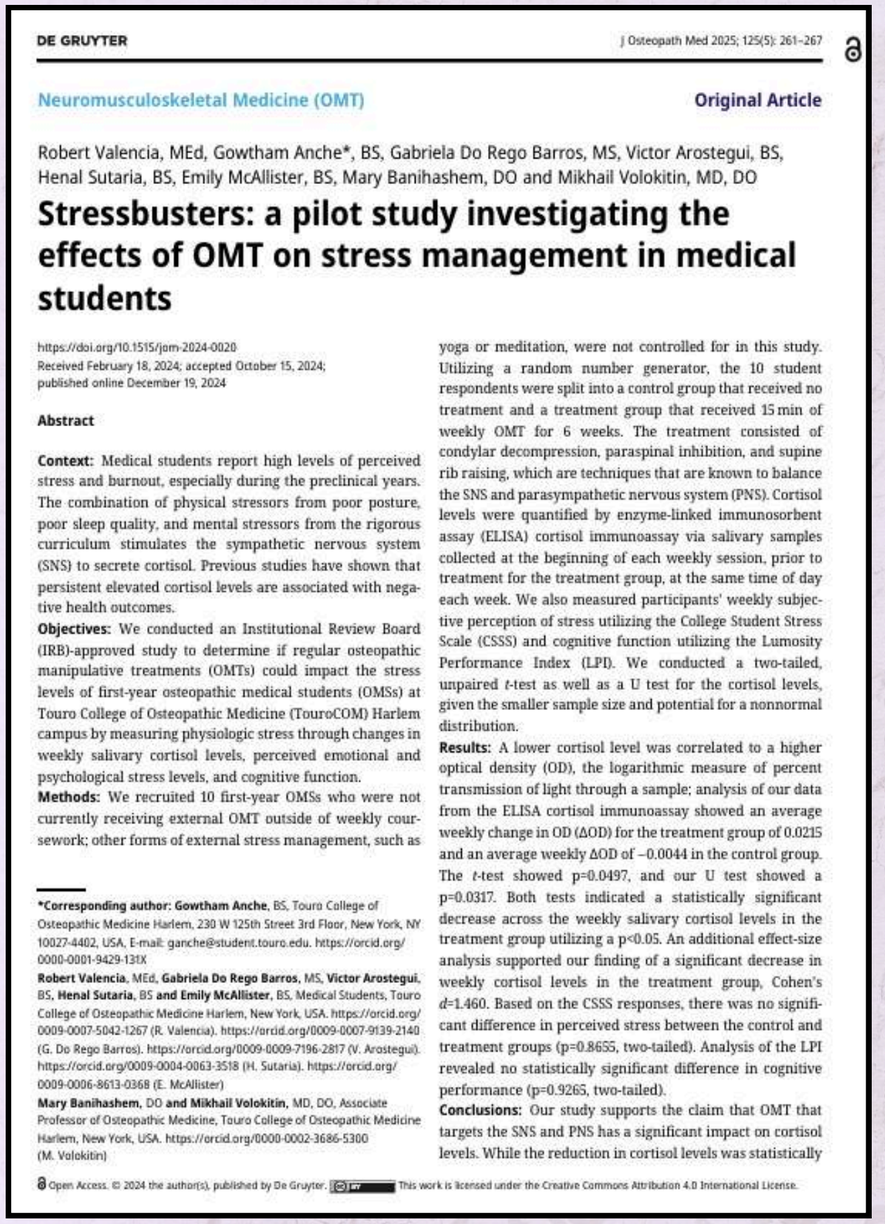


El régimen de D-TMO utilizado en el presente estudio produjo una disminución estadísticamente significativa en la fatiga autopercibida en los estudiantes de medicina osteopática.



El presente estudio piloto aleatorizado y controlado demostró que es factible evaluar el efecto del TMO sobre los síntomas de fatiga, estrés y depresión en estudiantes de medicina osteopática de primer año.

ARTÍCULO 3



DE GRUYTER J Osteopath Med 2025; 125(5): 261–267

Neuromusculoskeletal Medicine (OMT) Original Article

Robert Valencia, MEd, Gowtham Anche*, BS, Gabriela Do Rego Barros, MS, Victor Arostegui, BS, Henal Sutaria, BS, Emily McAllister, BS, Mary Banihashem, DO and Mikhail Volokitin, MD, DO

Stressbusters: a pilot study investigating the effects of OMT on stress management in medical students

https://doi.org/10.1515/jom-2024-0020
Received February 18, 2024; accepted October 15, 2024;
published online December 19, 2024

Abstract

Context: Medical students report high levels of perceived stress and burnout, especially during the preclinical years. The combination of physical stressors from poor posture, poor sleep quality, and mental stressors from the rigorous curriculum stimulates the sympathetic nervous system (SNS) to secrete cortisol. Previous studies have shown that persistent elevated cortisol levels are associated with negative health outcomes.

Objectives: We conducted an Institutional Review Board (IRB)-approved study to determine if regular osteopathic manipulative treatments (OMTs) could impact the stress levels of first-year osteopathic medical students (OMSs) at Touro College of Osteopathic Medicine (TouroCOM) Harlem campus by measuring physiologic stress through changes in weekly salivary cortisol levels, perceived emotional and psychological stress levels, and cognitive function.

Methods: We recruited 10 first-year OMSs who were not currently receiving external OMT outside of weekly coursework; other forms of external stress management, such as yoga or meditation, were not controlled for in this study. Utilizing a random number generator, the 10 student respondents were split into a control group that received no treatment and a treatment group that received 15 min of weekly OMT for 6 weeks. The treatment consisted of condylar decompression, paraspinal inhibition, and supine rib raising, which are techniques that are known to balance the SNS and parasympathetic nervous system (PNS). Cortisol levels were quantified by enzyme-linked immunosorbent assay (ELISA) cortisol immunoassay via salivary samples collected at the beginning of each weekly session, prior to treatment for the treatment group, at the same time of day each week. We also measured participants' weekly subjective perception of stress utilizing the College Student Stress Scale (CSSS) and cognitive function utilizing the Lumosity Performance Index (LPI). We conducted a two-tailed, unpaired *t*-test as well as a U test for the cortisol levels, given the smaller sample size and potential for a nonnormal distribution.

Results: A lower cortisol level was correlated to a higher optical density (OD), the logarithmic measure of percent transmission of light through a sample; analysis of our data from the ELISA cortisol immunoassay showed an average weekly change in OD (ΔOD) for the treatment group of 0.0215 and an average weekly ΔOD of −0.0044 in the control group. The *t*-test showed $p=0.0497$, and our U test showed a $p=0.0317$. Both tests indicated a statistically significant decrease across the weekly salivary cortisol levels in the treatment group utilizing a $p<0.05$. An additional effect-size analysis supported our finding of a significant decrease in weekly cortisol levels in the treatment group, Cohen's $d=1.460$. Based on the CSSS responses, there was no significant difference in perceived stress between the control and treatment groups ($p=0.8655$, two-tailed). Analysis of the LPI revealed no statistically significant difference in cognitive performance ($p=0.9265$, two-tailed).

Conclusions: Our study supports the claim that OMT that targets the SNS and PNS has a significant impact on cortisol levels. While the reduction in cortisol levels was statistically

***Corresponding author: Gowtham Anche**, BS, Touro College of Osteopathic Medicine Harlem, 230 W 125th Street 3rd Floor, New York, NY 10027-4402, USA, E-mail: ganche@student.touro.edu. https://orcid.org/0000-0001-9429-131X

Robert Valencia, MEd, **Gabriela Do Rego Barros**, MS, **Victor Arostegui**, BS, **Henal Sutaria**, BS **and Emily McAllister**, BS, Medical Students, Touro College of Osteopathic Medicine Harlem, New York, USA. https://orcid.org/0009-0007-5042-1267 (R. Valencia). https://orcid.org/0009-0007-9139-2140 (G. Do Rego Barros). https://orcid.org/0009-0009-7196-2817 (V. Arostegui). https://orcid.org/0009-0004-0063-3518 (H. Sutaria). https://orcid.org/0009-0006-8613-0368 (E. McAllister)

Mary Banihashem, DO **and Mikhail Volokitin**, MD, DO, Associate Professor of Osteopathic Medicine, Touro College of Osteopathic Medicine Harlem, New York, USA. https://orcid.org/0000-0002-3686-5300 (M. Volokitin)

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Informacion tecnica del articulo

- PMID: 39692236
- DOI: 10.1515/jom-2024-0020
- Tipo: Estudio piloto experimental, con grupo control y grupo de tratamiento

Cuartil y factor de impacto

- Corresponden al Journal of Osteopathic Medicine
- Está incluida en ESCI.
- Aparece como Q2 en algunos registros de JCR 2025
- En SCImago/Scopus figura como Q1 en determinadas categorías.
- JIF de 1,1
- CiteScore 2,6
- SJR 0,445.

Fecha de publiacion: 19 de diciembre de 2024

- **Objetivo:** Determinar si los tratamientos manipulativos osteopáticos regulares podrían afectar los niveles de estrés de los estudiantes de medicina.
- **Población:** Estudiantes de primer año de Medicina
- **Muestra:** Participaron 10 estudiantes (2 hombres y 8 mujeres, de entre 23 y 28 años).
- **Duracion:** 6 semanas
- **Distribución:** 5 estudiantes recibieron tratamiento manipulativo osteopático y 5 no recibió dicho tratamiento.
- **Tiempo de tratamiento:** 15 min por sesion (1 sesión semanal)

LIMITACIONES

La muestra fue muy pequeña solamente participaron 10 estudiantes

No se controlaron por completo otras actividades de manejo del estrés

No se midió directamente la intensidad del dolor musculoesquelético.

Se necesitan estudios más grandes para confirmar los resultados.

Table 1: Average weekly ΔOD of each participant.

Group	Participant ΔOD					Avg ΔOD
Treatment	0.05465	0.0062167	−0.00477	0.01327	0.038	0.02147334
Control	−0.0096	0.0052	−0.00515	−0.00297	−0.009483	−0.0044

ΔOD, change in optical density.

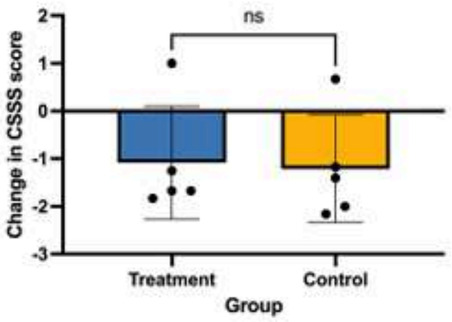


Figure 2: Average weekly changes in College Student Stress Scale (CSSS) score.

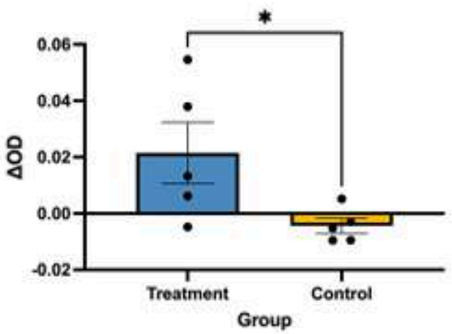


Figure 3: Average weekly changes in optical density (ΔOD).

0.01; range, −0.005 to 0.05) with a standard deviation (SD) of

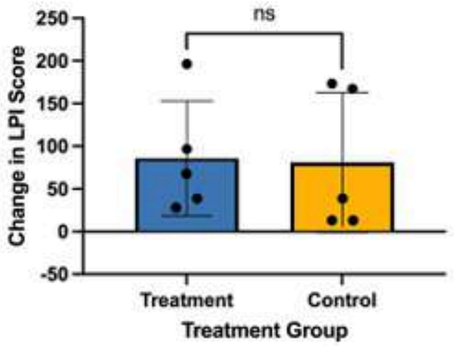


Figure 4: Average weekly change in Lumosity Performance Index (LPI).

control and treatment groups (p=0.9, two-tailed). The mean weekly change in perceived stress scores was −1.1 (median, −1.7; range, −1.8 to 1; SD=1.2) for the treatment group and a mean of −1.2 (median, −1.4; range, −2.2 to 0.7; SD=1.1) for the control group, with a negligible effect size (R^2=0.003). The CSSS score change for both groups is visualized in Figure 2.

A two-tailed *t*-test to compare the average ΔOD between the treatment and control groups. The *t*-test yielded a *t* value of 2.3 with 8 degrees of freedom (df), resulting in a p value of 0.049, which is statistically significant at the p<0.05 level, as shown in Figure 3. This test produced a p value of 0.03, further supporting statistical significance. The effect sizes were calculated utilizing the means and SDs. Cohen's *d* was found to be 1.5, and the effect-size *r* was 0.6. The power of this study is calculated to be 53 %.

Analysis of the change in LPI revealed no statistically

CONCLUSIÓN

La terapia de manipulación osteopática mostró una reducción estadísticamente significativa del cortisol salival en estudiantes de primer año de medicina osteopática. No obstante, no se encontraron diferencias significativas en el estrés percibido ni en la función cognitiva. Debido al tamaño reducido de la muestra y a que no se evaluó directamente el dolor muscular, se requieren investigaciones más amplias para determinar la magnitud y la relevancia clínica de sus beneficios.

La terapia de osteopática, aplicada durante 15 minutos semanales por 6 semanas, se asoció con una reducción estadísticamente significativa del cortisol salival en estudiantes de primer año de Medicina (p = 0,0317), esto sugiere un posible beneficio en la respuesta fisiológica al estrés.

Utiliza técnicas manuales para ayudar a aliviar ciertas molestias musculoesqueléticas y mejorar la movilidad, por lo que puede considerarse un complemento en el manejo del dolor físico.

Respuesta PICO

En estudiantes de primer año de medicina de la Universidad Tecnológica del Perú con mialgia asociada al estrés académico (P), la terapia osteopática (I), frente a la ausencia de tratamiento (C), logra una reducción significativa de la intensidad del dolor musculoesquelético (O).

Muchas gracias